

LESSON 5-6 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2

Drawing Nets of Solids

Maryland Standard 6.G.A.4 • Learning target – I can draw the net of a solid by unfolding its faces onto a flat surface.

REMEMBER

- A **net** is a flat pattern that folds into a solid. Each face appears **once**, with no overlaps.
- Rectangular prism**: 6 rectangles. **Square pyramid**: 1 square + 4 triangles. **Triangular prism**: 2 triangles + 3 rectangles.

WORKED EXAMPLE

Plan a net for a box 5 cm × 3 cm × 2 cm.

- Step 1** Faces come in pairs: 5 × 3, 5 × 2, 3 × 2
- Step 2** Draw the 5 × 3 base. Attach a 5 × 2 face to each long edge.
- Step 3** Attach a 3 × 2 face to each short edge. Add the 5 × 3 top to one 5 × 2 face.

Answer 6 rectangles, two of each size

WORD BANK

net – a flat pattern that folds into a solid

face – a flat side of a solid

rectangular prism – a box shape

square pyramid – a square base + 4 triangles

PRACTICE

show your work • label units

1. Draw a net for this box. Label the length and width of every face.

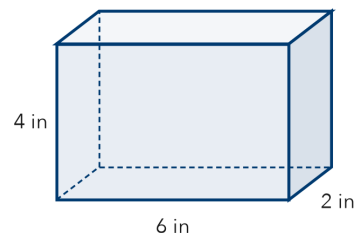
Step 1 Draw the bottom face first.

Step 2 Attach a face to each edge of the bottom.

Step 3 Add the top. Label every face.

Draw the net here.

Face sizes _____



2. How many of each shape are in the net?

a. cube: squares ____

b. square pyramid: triangles ____

c. triangular prism: rectangles ____

d. pentagonal pyramid: triangles ____

3. The net of this square pyramid has a square and 4 triangles.

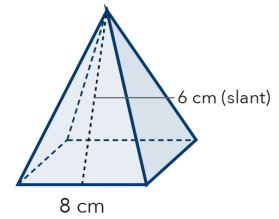
Step 1 Square: side $\times$ side.

Step 2 Triangle: $\frac{1}{2} \times \text{base} \times \text{height}$.

a. Area of the square _____ cm^2

b. Area of one triangle _____ cm^2

All 4 triangles _____ cm^2



4. **Cereal Box.** A cereal box is 8 in long, 3 in wide, and 12 in tall. Its net has **3 pairs** of matching faces. List the size of each pair.

Face pairs _____

5. **Will It Fold?** Jamal draws 6 squares in one long row and says it is a net for a cube. Is he right? Explain.

Frame Jamal is ____ because a row of 6 squares ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Open an empty cereal box along its edges. Show someone its net.

LESSON 5-6 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Reading Nets and Solids**

Maryland Standard 6.G.A.4 • Learning target – I can name the solid a net makes by reading the shapes in the net.

REMEMBER

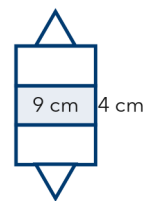
- **Prism:** 2 matching bases + rectangles. **Pyramid:** 1 base + triangles that meet at a point.
- The **base shape** names the solid (triangular, square, pentagonal ...).

WORKED EXAMPLE

A net has 1 hexagon and 6 triangles.

Step 1 One base with triangles around it means a pyramid.**Step 2** The base is a hexagon.**Answer** hexagonal pyramid**WORD BANK****prism** – 2 matching bases + rectangles**pyramid** – 1 base + triangles that meet at a point**base shape** – the shape that names the solid**edge** – where two faces meet**PRACTICE***show your work • label units*

1. Look at the net.

Step 1 Count the triangles and the rectangles.**Step 2** Two matching triangles + rectangles make a prism.**a.** Solid: _____**b.** Number of faces: ____

2. Name the solid each net makes.

Step 1 Two matching shapes: prism. One base + triangles: pyramid.**Step 2** Use the base shape in the name.**a.** 1 rectangle + 4 triangles: _____**b.** 2 hexagons + 6 rectangles: _____**c.** 2 triangles + 3 rectangles: _____**d.** 1 pentagon + 5 triangles: _____**e.** 6 squares: _____**f.** 2 pentagons + 5 rectangles: _____

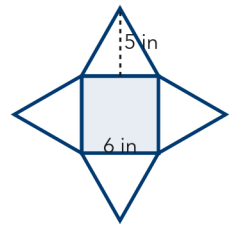
3. This net folds into a square pyramid.

Step 1 Base: 6×6 .

Step 2 One triangle: $\frac{1}{2} \times 6 \times 5$. Then multiply by 4.

a. Area of the base: _____ in²

b. Area of all 4 triangles: _____ in²



4. **Mystery Box.** A net has 2 **squares** (4 cm on each side) and 4 **rectangles** (4 cm $\times$ 9 cm). Name the solid. Give its length, width, and height.

Answer _____

TEST PRACTICE

choose the best answer

5. A triangular prism has 5 faces. How many edges does it have?

Step 1 Count the edges on each triangle end.

Step 2 Add the edges that join the two ends.

A 5 **B** 6 **C** 9 **D** 12

6. How many faces does a hexagonal pyramid have?

A 6 **B** 7 **C** 8 **D** 12

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Find a box at home. Tell someone what shapes its net would have and how many of each.